

Mastering Sets, Relations, and Applications

A Deep Dive into Mathematical Foundations

Comprehensive Learning Module

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1 The Theory of Sets (Ensembles)

1.1 Introduction to Sets

Mathematics is built upon the concept of a **Set**. Before we look at numbers or equations, we must understand how to group objects together.

DEFINITION: Set

A **set** is a well-defined collection of distinct objects. The objects contained in a set are called **elements** (or members).

INTUITION: The "Bag" Analogy

Imagine a set as a transparent bag.

- You can put things inside (elements).
- The order in which you put them in does not matter. A bag containing an apple and a pear is the same as a bag containing a pear and an apple.
- Repetition does not matter. If you put the *exact same* object in twice, the set still only "contains" it once.

1.1.1 Notation

We generally use uppercase letters for sets (e.g., A, B, E) and lowercase letters for elements (e.g., x, y, a).

- **Membership ($\in$):** If x is inside set A , we write $x \in A$.
- **Non-membership ($\notin$):** If x is not inside set A , we write $x \notin A$.

1.2 Describing Sets

There are two standard ways to write down what is inside a set.

1.2.1 1. Extension (Roster Method)

We list every single element inside curly braces $\{\}$.

$$E = \{1, 2, 3\}$$

This is useful for small sets but impossible for infinite sets (like all numbers).

1.2.2 2. Comprehension (Set-Builder Notation)

We describe the rule or property that elements must satisfy.

$$E = \{x \in \Omega \mid P(x)\}$$

How to read this: "The set of all elements x belonging to Ω **such that** the property $P(x)$ is true."

EXAMPLE: Converting Notation

Description: The set of even integers.

- **Roster (Partial):** $E = \{\dots, -4, -2, 0, 2, 4, \dots\}$
- **Builder:** $E = \{x \in \mathbb{Z} \mid \exists k \in \mathbb{Z}, x = 2k\}$

1.3 Subsets and Power Sets

1.3.1 Inclusion ($\subset$)

One of the most common logical operations is determining if one set fits entirely inside another.

DEFINITION: Subset

Set A is a subset of set B (written $A \subset B$ or $A \subseteq B$) if every element of A is also an element of B .

$$A \subset B \iff (\forall x, x \in A \implies x \in B)$$

CAUTION: Element vs. Subset

This is a critical distinction:

- $x \in A$: x is an **item** inside the bag.
- $\{x\} \subset A$: $\{x\}$ is a **smaller bag** containing x , which is inside the big bag A .

Example: Let $A = \{1, 2\}$. $1 \in A$ is True. $\{1\} \subset A$ is True. $1 \subset A$ is **False** (numbers are not sets).

1.3.2 The Power Set ($\mathcal{P}(E)$)

The Power Set is the "Set of all Subsets." It is a meta-set. If you take every possible combination of elements from set E and put them into new sets, you form $\mathcal{P}(E)$.

- If E has n elements, $\mathcal{P}(E)$ contains 2^n elements.

- The empty set $\emptyset$ and the set E itself are always inside $\mathcal{P}(E)$.

EXAMPLE: Calculating a Power Set

Let $E = \{a, b, c\}$. The size is $n = 3$. We expect $2^3 = 8$ subsets.

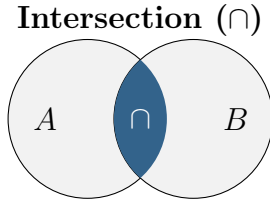
1. **0 elements:** $\emptyset$
2. **1 element:** $\{a\}, \{b\}, \{c\}$
3. **2 elements:** $\{a, b\}, \{a, c\}, \{b, c\}$
4. **3 elements:** $\{a, b, c\}$

$$\mathcal{P}(E) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

2 Operations on Sets

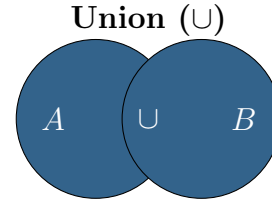
We perform arithmetic on numbers ($3 + 5 = 8$). Similarly, we perform operations on sets to create new sets.

2.1 Union and Intersection



Logic: AND. Elements in A **and** B .

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$



Logic: OR. Elements in A **or** B (or both).

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

2.2 Difference and Complements

2.2.1 Set Difference ($A \setminus B$)

Often written as $A - B$. This is "Subtraction" for sets. It means: "Everything in A , **excluding** anything that is also in B ."

$$A \setminus B = \{x \in A \mid x \notin B\}$$

2.2.2 The Complement ($\bar{A}$ or A^c)

The complement requires a "Universal Set" E . $\bar{A}$ is everything in the universe that is **not** in A .

$$\bar{A} = E \setminus A$$

Property: $\bar{\bar{A}} = A$ (The complement of a complement is the original set).

2.3 De Morgan's Laws

These are powerful rules for how complements interact with unions and intersections. They essentially say: "To break a bar over a group, you must flip the symbol."

De Morgan's Laws For any sets A and B :

$$1. \overline{A \cup B} = \bar{A} \cap \bar{B} \quad (\text{The NOT of (A or B) is (NOT A and NOT B)})$$

$$2. \overline{A \cap B} = \bar{A} \cup \bar{B} \quad (\text{The NOT of (A and B) is (NOT A or NOT B)})$$

2.4 Cartesian Product ($A \times B$)

This operation creates a set of pairs. It changes the nature of the object; we go from single points to coordinates.

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

If A has n elements and B has m elements, $A \times B$ has $n \times m$ elements.

3 Relations

3.1 Defining a Relation

In math, a relation describes a link between elements.

DEFINITION: Binary Relation

A binary relation $\mathcal{R}$ on a set E is a subset of the Cartesian product $E \times E$. If the pair $(x, y) \in \mathcal{R}$, we write $x\mathcal{R}y$.

3.2 Properties of Relations

Relations are classified by how they behave. Let's define these properties using a relation $\mathcal{R}$ on set E .

1. **Reflexivity (The Mirror Property)** Every element is related to itself.

$$\forall x \in E, \quad x\mathcal{R}x$$

Example: "is equal to" ($=$) is reflexive. "is greater than" ($>$) is **not** reflexive (5 is not greater than 5).

2. **Symmetry (The Two-Way Street)** If x relates to y , then y must relate to x .

$$\forall x, y \in E, \quad x\mathcal{R}y \implies y\mathcal{R}x$$

Example: "is a sibling of" is symmetric. "is the father of" is **not** symmetric.

3. **Antisymmetry (The Strict Hierarchy)** If x relates to y AND y relates to x , it implies x and y are actually the same thing.

$$\forall x, y \in E, \quad (x\mathcal{R}y \wedge y\mathcal{R}x) \implies x = y$$

Example: "is less than or equal to" ($\leq$) is antisymmetric. If $x \leq y$ and $y \leq x$, then $x = y$.

4. **Transitivity (The Chain Reaction)** If x relates to y , and y relates to z , then x automatically relates to z .

$$\forall x, y, z \in E, \quad (x\mathcal{R}y \wedge y\mathcal{R}z) \implies x\mathcal{R}z$$

Example: "is taller than." If A is taller than B, and B is taller than C, A is taller than C.

3.3 Types of Relations

3.3.1 Equivalence Relations ($\sim$)

A relation is an **Equivalence Relation** if it is Reflexive, Symmetric, and Transitive.

- **Purpose:** It groups objects that are "similar" in some specific way.
- **Equivalence Class:** The set of all elements related to x , denoted $\bar{x}$ or $[x]$.

3.3.2 Order Relations ($\leq$)

A relation is an **Order Relation** if it is Reflexive, Antisymmetric, and Transitive.

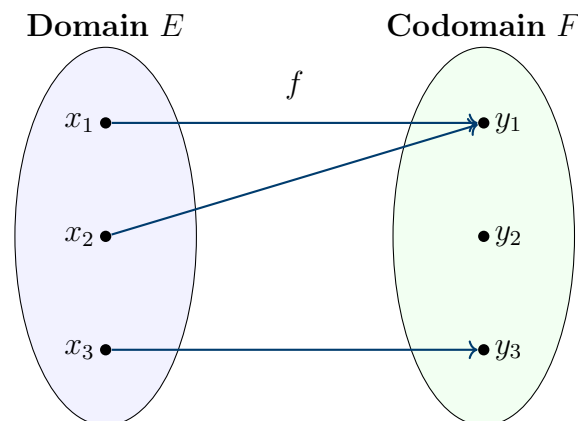
- **Purpose:** It arranges objects in a hierarchy.

4 Applications (Functions)

This is arguably the most important section for Calculus and Analysis.

4.1 Formal Definition

An application (function) $f : E \rightarrow F$ is a relation where **every** input $x \in E$ has **exactly one** output $y \in F$.



Note in the diagram: y_2 is not hit by any arrow. This is allowed for a general function.

4.2 Properties: Injection, Surjection, Bijection

INTUITION: The Arrow Target Metaphor

Think of the domain E as archers and the codomain F as targets.

- **Function Rule:** Every archer shoots exactly one arrow.
- **Injection:** No two archers hit the same target.
- **Surjection:** Every target gets hit at least once.
- **Bijection:** Perfect match. Every archer hits a unique target, and all targets are hit.

4.2.1 1. Injective (One-to-One)

$$\forall x_1, x_2 \in E, \quad f(x_1) = f(x_2) \implies x_1 = x_2$$

Proof Strategy: Assume $f(x_1) = f(x_2)$, do algebra, and try to prove $x_1 = x_2$.

4.2.2 2. Surjective (Onto)

$$\forall y \in F, \quad \exists x \in E \text{ such that } y = f(x)$$

Proof Strategy: Pick an arbitrary y . Can you solve the equation $y = f(x)$ for x ? Does that x exist in the domain?

4.2.3 3. Bijective

A function is bijective if it is both injective and surjective.

$$\forall y \in F, \quad \exists! x \in E \text{ such that } y = f(x)$$

($\exists!$ means "there exists **unique**").

4.3 Inverse Functions

Only bijective functions have an inverse $f^{-1} : F \rightarrow E$.

$$y = f(x) \iff x = f^{-1}(y)$$

EXAMPLE: Detailed Analysis of a Function

Let $f : \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R} \setminus \{2\}$ defined by $f(x) = \frac{2x+1}{x-1}$.

Step 1: Check Injectivity Assume $f(a) = f(b)$:

$$\frac{2a+1}{a-1} = \frac{2b+1}{b-1}$$

Cross multiply:

$$(2a+1)(b-1) = (2b+1)(a-1)$$

$$2ab - 2a + b - 1 = 2ab - 2b + a - 1$$

Cancel $2ab$ and -1 :

$$-2a + b = -2b + a$$

$$3b = 3a \implies a = b$$

Conclusion: f is Injective.

Step 2: Check Surjectivity Let $y \in \mathbb{R} \setminus \{2\}$. Solve $y = \frac{2x+1}{x-1}$ for x .

$$y(x-1) = 2x+1$$

$$xy - y = 2x + 1$$

$$xy - 2x = y + 1$$

$$x(y-2) = y+1$$

$$x = \frac{y+1}{y-2}$$

Is this valid? Yes, because we stated $y \neq 2$, so we never divide by zero. **Conclusion:** f is Surjective.

Step 3: Inverse Since f is bijective, $f^{-1}(y) = \frac{y+1}{y-2}$.

5 Practice Exercises

Level 1: Conceptual Understanding

Exercise 1. Let $A = \{1, 2, 3\}$ and $B = \{3, 4\}$.

1. Write down $A \cap B$ and $A \cup B$.
2. Write down $A \times B$. How many elements does it have?
3. Write down $\mathcal{P}(A \cap B)$.

Level 2: Relations Logic

Exercise 2. Consider the set $E = \mathbb{Z}$ (integers). Let $\mathcal{R}$ be the relation defined by:

$$x\mathcal{R}y \iff x - y \text{ is a multiple of } 3$$

(Mathematically: $\exists k \in \mathbb{Z}, x - y = 3k$).

1. Show that $\mathcal{R}$ is Reflexive. (Hint: is $x - x$ a multiple of 3?)
2. Show that $\mathcal{R}$ is Symmetric. (Hint: if $x - y = 3k$, what is $y - x$?)
3. Show that $\mathcal{R}$ is Transitive.

Level 3: Functions and Proofs

Exercise 3. Let $f : [1, +\infty[\rightarrow [0, +\infty[$ be defined by $f(x) = \sqrt{x-1}$.

1. Is f injective? Proof required.
2. Is f surjective? Proof required.
3. If it is bijective, find the expression for $f^{-1}(x)$.

Solutions

Solution 1:

1. $A \cap B = \{3\}$. $A \cup B = \{1, 2, 3, 4\}$.
2. $A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4), (3, 3), (3, 4)\}$. Size is $3 \times 2 = 6$.
3. $A \cap B = \{3\}$, so $\mathcal{P}(\{3\}) = \{\emptyset, \{3\}\}$.

Solution 2:

1. **Reflexive:** $x - x = 0$. Since $0 = 3 \times 0$, it is a multiple of 3. Yes.
2. **Symmetric:** If $x - y = 3k$, then $y - x = -(x - y) = -3k = 3(-k)$. Since $-k$ is an integer, this is a multiple of 3. Yes.
3. **Transitive:** If $x - y = 3k$ and $y - z = 3j$. Adding equations: $(x - y) + (y - z) = 3k + 3j \implies x - z = 3(k + j)$. Yes.

Solution 3:

1. **Injectivity:** Let $f(x_1) = f(x_2) \implies \sqrt{x_1 - 1} = \sqrt{x_2 - 1}$. Square both sides: $x_1 - 1 = x_2 - 1 \implies x_1 = x_2$. Yes.
2. **Surjectivity:** Let $y \in [0, +\infty[$. We want $y = \sqrt{x - 1}$. Square both sides: $y^2 = x - 1 \implies x = y^2 + 1$. Since $y \geq 0$, $y^2 \geq 0$, so $x \geq 1$. This x is in the domain. Yes.
3. **Inverse:** $f^{-1}(y) = y^2 + 1$.